EECS 221: Machine Learning for Science

Data–Compute Trade-offs in Neural PDE Forecasting

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Mid-Project Report

Project goal. Neural PDE forecasting models must be useful under limited data and compute. This project compares how accurately and efficiently three architecture families (CNextU-Net, TFNO and AViT) learn Rayleigh–Bénard convection when training data and GPU time are limited. This check-in covers the most constrained setting of the study: 10% of the training data and 1 or 4 hours of training on one A100 GPU; the 100% runs follow. The abstract's three milestones are met: the data pipeline is validated against The Well's released checkpoints (Section 3), budgets are fixed from pilot timings (Section 4), and the first 10% results are in Section 5, with the revised plan and the changes since the abstract in Section 7.

1 Problem and Data

Physical system. Rayleigh–Bénard convection is a horizontal layer of fluid heated from below and cooled from above. Once buoyancy overcomes the damping by viscosity and heat diffusion, the layer overturns into rising warm and sinking cold plumes with thin boundary layers along the walls: the classical model of convection in the atmosphere, the oceans and stars (Fig. 3, Appendix A). The data solve the two-dimensional Boussinesq equations for buoyancy b, velocity (u, w) and pressure p in a 4×1 box, periodic in x, between no-slip walls held at $b = 1$ (bottom) and $b = 0$ (top). Two numbers set a regime: the *Rayleigh number* $\mathrm{Ra} = g\alpha\,\Delta T\,H^3/(\nu\kappa)$, the buoyancy forcing relative to viscous and thermal damping (10^6 to 10^{10}; higher means more turbulent flow with finer plumes), and the *Prandtl number* $\mathrm{Pr} = \nu/\kappa$, momentum relative to heat diffusion (0.1 to 10). Each trajectory starts from rest with a small random perturbation, so convection first has to grow (*onset*) before it becomes statistically steady (*developed* convection).

Data. `rayleigh_benard` from The Well [1, 13]: 1,750 trajectories, 50 in each of 35 (Ra, Pr) regimes, 367 GB. Each has 200 frames, 0.25 time units apart, of the four fields on a 512×128 grid (uniform in x, Chebyshev nodes in y). The official split assigns whole trajectories: 1,400 for training, 175 for validation and 175 for testing (40 / 5 / 5 per regime). We train on a nested 10% subset (140 trajectories, four per regime) and evaluate on the full test split. Every choice made before testing (learning rate, monitoring, the onset cut, rollout start frames) used validation data only.

Task. From four consecutive frames of all four fields, a model predicts the next frame; longer forecasts feed its predictions back in (autoregressive rollout).

2 Method and Baselines

Models (Table 1). The Well's implementations of three families, trained from random initialisation with matched parameter counts. Our TFNO uses the current code; the paper-era code skipped TFNO's Fourier layers (Section 3).

Model	Family	Parameters	Normalisation	Relation to released checkpoint
CNextU-Net [2]	ConvNeXt U-Net, 4 stages	18.58M	LayerNorm	same architecture
TFNO [5, 3]	Fourier operator, 16×16 modes	19.28M	none	same configuration, fixed forward
AViT [4]	axial-attention ViT	18.57M	GroupNorm, LayerNorm	none released; width 384 → 432

Table 1: The three model families.

Training. One recipe for every model: AdamW, one-step MSE on normalised fields, batch 64, fp32, peak learning rate 10^{-3} with 500 warm-up steps and cosine decay over 4 hours of active GPU training on one A100, and two seeds. The 1-hour results are these runs' checkpoints after one hour. Details, and how the recipe differs from the paper's, are in Appendix B.

Baselines. *Persistence* copies the last input frame. It costs nothing to train, so a model is worth training only where it beats persistence. *The released CNextU-Net*, trained by The Well on all the data for 12 hours on an H100, is the full-data reference. Planned for the final report are a numerical-solver baseline (Dedalus on coarser grids, compared at equal accuracy or equal cost [15]) and baselines that need no GPU training: linear extrapolation in time, climatology and dynamic mode decomposition (Appendix A).

Metrics. VRMSE is the RMSE of a field divided by the target field's spatial standard deviation (The Well's metric; Appendix C), averaged over the four fields. We compute it on the fields as stored (*physical-field VRMSE*) and, to compare with the paper, on normalised fields (*paper protocol*); the two differ only while the velocity is nearly zero. One-step errors are averaged over all 196 windows of each of the 175 test trajectories and also split into *onset* (first input frame < 50) and *developed* windows. Rollouts run 30 steps from frames 20, 80 and 140; the *failure horizon* is the first step at which VRMSE exceeds 1, 2 or 5 (a constant field at the target's mean scores about 1).

3 Milestone 1: Pipeline Validated Against the Released Checkpoints

Scored with our data path and metric on every window of the 175 test trajectories, the released CNextU-Net reproduces the paper's test VRMSE and its rollout range (Table 2). The released TFNO reproduces its value too (0.663 against 0.657), but only with the forward pass it was trained with: a bug in the paper-era code skipped its Fourier layers, so it is effectively a per-pixel network that tracks persistence (Appendix D). The check also showed that the all-window mean is dominated by the first ∼50 frames, where the fluid is nearly at rest and relative velocity errors are large: from frame 50 on, the released CNextU-Net scores 0.038, against 0.224 over all windows (Fig. 4, Appendix E). We therefore report onset and developed windows separately.

	Paper-protocol one-step VRMSE		Rollout from frame 0, steps 6–12 / 13–30	
Model	Ours (released weights)	Published	Ours	Published
CNextU-Net	0.669	0.670	36 / 35	> 10 / > 10

Table 2: The released CNextU-Net scored by our pipeline against the paper's Tables 2 and 3 (test split). The released TFNO is in Appendix D (Table 8).

4 Milestone 2: Budgets Fixed from Pilot Timings

Budget clock. The cluster's shared file system delivers at most ∼98 training samples per second into one node and stalls for minutes at a time, while TFNO and AViT consume 116 and 330 per second of GPU time. A wall-clock budget would partly measure our storage, so the 1- and 4-hour budgets count *active GPU training time*; allocated GPU time, data waits and SU are reported separately (Appendix F).

Learning rate. The paper's rates ($5 \cdot 10^{-4}$ for CNextU-Net, 10^{-4} for TFNO) were chosen for 12-hour runs. An equal-budget search on validation data (20 minutes of training per run) chose 10^{-3} for all three models, clearly better than the paper's rates at this budget (CNextU-Net 0.408 against 0.469, TFNO 0.302 against 0.603). It is the largest rate tried, so it is the best tested rather than an established optimum (Appendix B).

Batch size and precision. Batch 64 (The Well's) fits every model in fp32 on an 80 GB A100 (CNextU-Net peaks at 61 GB). TFNO's Fourier layers require fp32, so all models train in fp32. This favours TFNO: bf16 would speed up a training step 1.7 times for CNextU-Net and 2.2 times for AViT.

5 Milestone 3: First Results with 10% of the Data

Training curves (Fig. 1). All six runs completed their 4 hours. AViT and CNextU-Net reach similar validation error, TFNO plateaus after ∼1.5 hours, and CNextU-Net's validation error peaks near one hour, while its learning rate is still high.

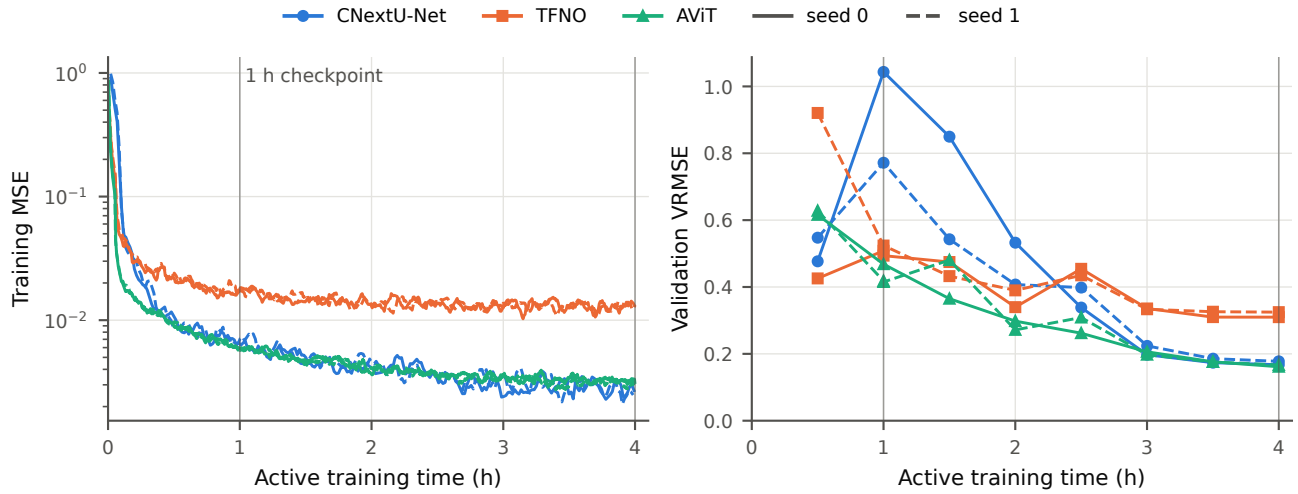


Figure 1: Training MSE in normalised units (left; smoothed over 3 minutes of training) and validation VRMSE on the 64 fixed windows, physical fields (right), against active training time, 10% data. Solid: seed 0; dashed: seed 1. Vertical lines: the 1-hour and 4-hour checkpoints.

One-step error (Table 3). After 4 hours CNextU-Net is the most accurate in developed convection (0.050, against 0.075 for AViT and 0.109 for TFNO) and AViT during onset; over all windows the two are level. The spread between seeds is small at 4 hours (standard deviation at most 0.006). The released CNextU-Net, with all the data and a longer, different schedule, remains better in developed convection (0.038). Persistence beats every model during onset, a velocity effect (on buoyancy all three beat it; Appendix G), and is 2–4.5 times worse in developed convection.

Rollouts (Table 4). One-step accuracy does not carry over: a model trained on one-step targets sees its own errors as inputs in a rollout. CNextU-Net is the most stable: its 30-step errors from frames 80 and 140 (0.53, 0.40) are below persistence's (0.86, 0.54), and 89% of its rollouts stay below VRMSE 1

Model (10% data, checkpoint)	Physical-field one-step VRMSE			Paper protocol
	all windows	onset	developed	(all windows)
CNextU-Net, 4 h	0.136 ± 0.006	0.387 ± 0.025	**0.050** ± 0.000	0.349 ± 0.022
TFNO, 4 h	0.209 ± 0.001	0.502 ± 0.003	0.109 ± 0.000	0.488 ± 0.001
AViT, 4 h	**0.134** ± 0.001	**0.308** ± 0.003	0.075 ± 0.000	**0.298** ± 0.002
CNextU-Net, 1 h	0.728 ± 0.176	2.618 ± 0.681	**0.081** ± 0.003	2.345 ± 0.600
TFNO, 1 h	0.473 ± 0.078	1.505 ± 0.305	0.120 ± 0.000	1.362 ± 0.249
AViT, 1 h	**0.394** ± 0.036	**1.299** ± 0.140	0.085 ± 0.000	**1.171** ± 0.125
Released CNextU-Net (100%, paper)	0.224	0.766	0.038	0.669
Persistence	0.199	0.121	0.226	0.202

Table 3: One-step test error: mean ± standard deviation over the two seeds (0.000: below 0.0005). Onset: first input frame < 50; developed: ≥ 50. Bold: best of our models at each budget.

throughout. AViT degrades fastest and after 30 steps from developed flow is no better than persistence; TFNO lies in between. Under the paper's frame-0 protocol all three are in its "> 10" range.

Model	VRMSE after 30 steps from frame 20 / 80 / 140	Fraction staying below threshold through step 30: 1 / 2 / 5	Paper protocol, steps 6–12 / 13–30 (normalised)
CNextU-Net, 4 h	1.78 / **0.53** / **0.40**	**0.89** / **0.94** / **0.98**	**15** / **30**
TFNO, 4 h	1.79 / 0.59 / 0.45	0.75 / 0.85 / 0.91	51 / 77
AViT, 4 h	5.42 / 0.80 / 0.65	0.63 / 0.80 / 0.87	75 / 167
CNextU-Net, 1 h	3.60 / 1.09 / 1.04	0.34 / 0.74 / 0.79	202 / 148
TFNO, 1 h	**2.67** / **0.90** / **0.78**	**0.60** / **0.77** / **0.86**	**81** / **116**
AViT, 1 h	4.67 / 1.11 / 0.98	0.34 / 0.74 / 0.83	129 / 184
Released CNextU-Net (100%)			36 / 35
Persistence	0.68 / 0.86 / 0.54	0.94 / 1.00 / 1.00	0.27 / 0.50

Table 4: Rollouts on the test set, physical-field VRMSE unless stated, mean over the two seeds. Middle column: fraction of the 525 rollouts (175 trajectories × 3 start frames) whose error stays below each threshold at every step through step 30. Right: the paper's Table 3 protocol (published: > 10 in both windows); bold: best of our models at each budget.

With one hour of training the ranking changes: AViT has the lowest one-step error (0.39, against 0.47 for TFNO and 0.73 for CNextU-Net), TFNO the most stable rollouts (60% stay below VRMSE 1, against 34%), and CNextU-Net varies most between seeds.

Example predictions (Fig. 2). At Ra $= 10^8$ CNextU-Net follows the true plumes through 30 steps, TFNO blurs them and builds a spurious warm layer along the bottom wall, and AViT shows patch-shaped artifacts by step 12.

Summary. Further results are in the appendices: error by field and kinetic-energy spectra (Appendix G), cost to train and to use (Appendix H), and error by Rayleigh number and heat transport (Appendix I). Together with the results above:

- **Accuracy.** After 4 hours CNextU-Net is the most accurate and stable forecaster of developed convection, with 1.3 times the error of the released full-data model. TFNO's predictions are too smooth, and AViT's degrade fastest over a rollout.
- **Training time.** CNextU-Net gains most from longer training: from one to four hours its developed-convection error falls by 38%, against 12% for AViT and 9% for TFNO.

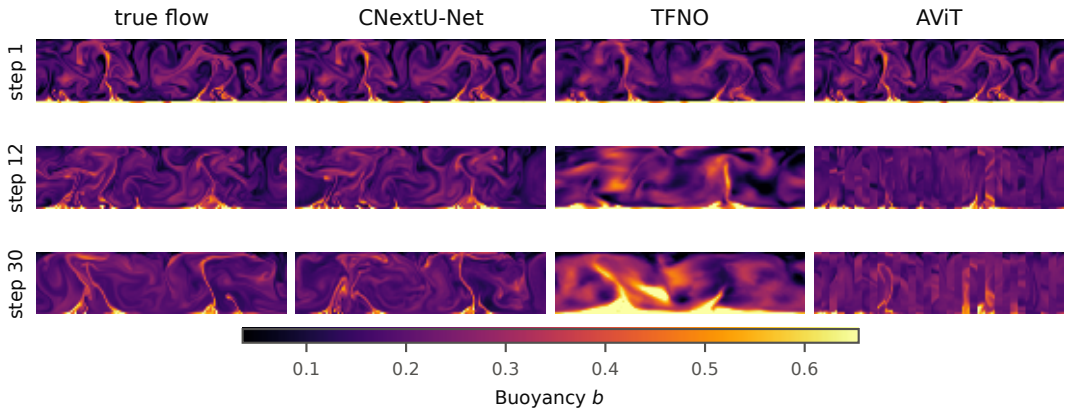


Figure 2: Buoyancy on a test trajectory at Ra = 10^8, Pr = 1, rolled out from frame 80: the true flow and our 10% / 4-hour models (seed 0) after 1, 12 and 30 steps. Bright: warm; dark: cold. One colour range is used for all panels: the 1st to 99th percentile of the true field away from the walls, so the hot bottom wall saturates.

- **Cost.** AViT takes six times as many optimizer steps as CNextU-Net with a seventh of its memory, but its runs were data-bound (11–12 hours of GPU time held for 4 hours of training). For single forecasts TFNO is the fastest (135 ms for 30 steps, against 163 ms for AViT and 249 ms for CNextU-Net); for batched forecasts AViT is the cheapest per sample.
- **Negative results.** No model beats persistence on velocity during onset, AViT is no better than persistence after 30 steps, and over a 30-step rollout only TFNO predicts the averaged heat transport better than persistence.
- **Caveats.** Two seeds on one fixed 10% subset; fp32 favours TFNO; the learning rate is at the edge of the tested grid.

6 Issues

Nothing blocks the study: every planned 10% run and evaluation finished. The open issues, mostly about reading data on HPC3, are in Table 5. All issues, including those already resolved, and where each run's GPU time went are in Appendix J.

7 Revised Plan

The core comparison is unchanged: three families, 10% and 100% data, two seeds, and checkpoints at one and four hours; so are the extensions and their order.

Changes since the abstract.

- Budgets count active GPU training time, not wall time (Section 4).
- The 48 hours of active training are unchanged, but the estimate is about 135 allocated GPU-hours instead of 100, the difference being waits for data; 72.6 are used so far (54.2 on the paid partition, billed 2,160 SU, and 18.3 on the free one).
- Persistence is added as a baseline, and a coarse-grid solver baseline is planned (Section 2).
- The released TFNO turned out to be affected by a bug, so only the released CNextU-Net is a valid reference.

Issue	Effect so far	Status
Shared file system (`/pub`): at most ∼98 samples/s per node; stalls of 5–22 minutes on all nodes	TFNO and AViT waited 2.5–2.8 h and 6.1–7.3 h for data per 4-hour run	**Open.** Test node-local staging before the 100% runs
File-system errors (`Errno 70`)	Crashed both AViT runs at once	**Mitigated.** Runs resume from checkpoints
Free GPU partition preempts jobs and is often full	Three 10% runs preempted	**Worked around.** Paid partition
Queue waits on the paid partition	Median about 3 h, up to 12–15 h	**Open.** Plan a day of slack
Allocated time is billed, including data waits	2,160 SU so far; the 100% runs need about 2,400 more	**Open.** Falls if staging removes the waits
TFNO cannot train in bf16	All models train in fp32, which favours TFNO	**Accepted**; stated as a caveat
The learning rate 10^{-3} is the largest tried	The optimum may be higher	**Accepted**; stated as a caveat
Models are trained on one-step targets only	AViT's rollouts no better than persistence; heat transport off over rollouts	**Open.** Outside the fixed recipe

Table 5: Open issues and those worked around or accepted. All issues: Table 16 (Appendix J).

- Learning rate 10^{-3} and fp32 for every model (Section 4).
- Evaluation: onset and developed convection reported separately, the paper's protocol alongside ours, failure thresholds 1, 2 and 5 instead of one, and error by Rayleigh number and heat transport added.

Next steps, in priority order. (1) Make data loading robust and test node-local staging. (2) Run the six 100% runs (about 2,400 SU) and compare 10% with 100% data at 1 and 4 hours, with accuracy–cost curves for both. (3) Run the solver baseline on CPUs. (4) Complete the analyses for the 100% runs (fields, spectra, Rayleigh number, heat transport, snapshots) and the reproducible final report. Only after the core: the abstract's extensions (input-noise robustness, a third seed, a 30% fraction, ConvLSTM). Optional, if the budget allows: 24-hour runs with the paper's learning rates (60–90 GPU-hours), a second draw of the 10% subset (about 25 GPU-hours), and batched inference timing.

Appendix A: Physical System, Data and Baselines in Detail

The full descriptions behind Sections 1 and 2.

Physical system (Fig. 3). Rayleigh–Bénard convection is a horizontal layer of fluid heated from below and cooled from above. Warm fluid near the bottom is lighter than the cold fluid above it; once this buoyancy forcing is strong enough to overcome the damping by viscosity and heat diffusion, the layer overturns into rising warm and sinking cold plumes, with thin thermal boundary layers along both walls. It is the classical model of buoyancy-driven convection, as in the atmosphere, the oceans, stellar interiors and the Earth's mantle, and a standard test problem for turbulence and heat transfer. The data solve the two-dimensional Boussinesq equations for buoyancy b (the temperature variable), velocity (u, w) and pressure p in a 4×1 box, periodic in x, between no-slip walls held at $b = 1$ (bottom) and $b = 0$ (top) [13]. Two dimensionless numbers set a regime:

- the *Rayleigh number* $\mathrm{Ra} = g\alpha\,\Delta T\,H^3/(\nu\kappa)$, the buoyancy forcing relative to viscous and thermal damping (layer height H, temperature difference ΔT, thermal expansion α, viscosity ν, thermal diffusivity κ). A larger Rayleigh number means stronger, more turbulent convection with finer plumes and thinner boundary layers (Fig. 3). The data span $\mathrm{Ra} = 10^6, 10^7, \ldots, 10^{10}$;
- the *Prandtl number* $\mathrm{Pr} = \nu/\kappa$, how fast momentum diffuses relative to heat (about 0.7 for air and 7 for water). The data span $\mathrm{Pr} = 0.1, 0.2, 0.5, 1, 2, 5, 10$.

In the data's units $\kappa = (\mathrm{Ra}\,\mathrm{Pr})^{-1/2}$ and $\nu = (\mathrm{Ra}/\mathrm{Pr})^{-1/2}$. Each trajectory starts from rest, with a linear (purely conductive) buoyancy profile plus a small random perturbation. Convection first has to grow out of this nearly still state (*onset*) and then settles into statistically steady, *developed* convection.

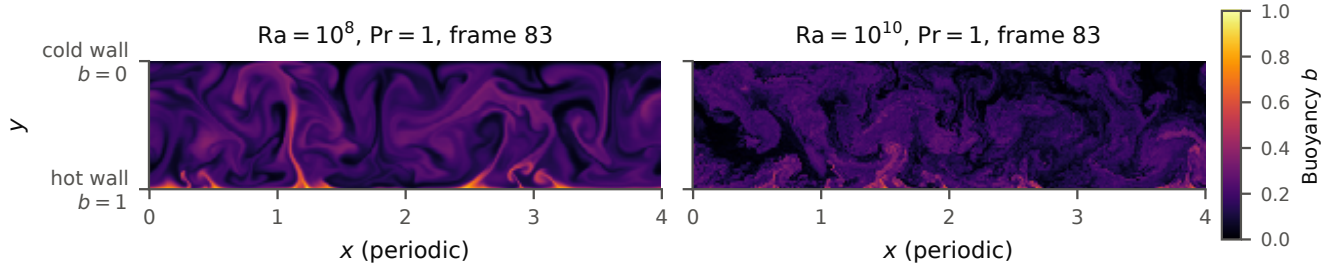


Figure 3: The physical system: true buoyancy on two test trajectories in developed convection, at two Rayleigh numbers. Warm fluid rises from the hot bottom wall in plumes, and the flow becomes much finer at the higher Rayleigh number. The grid is uniform in x and clustered towards the walls in y.

Data. `rayleigh_benard` from The Well [1, 13]: 1,750 trajectories, 50 in each of the 35 (Rayleigh, Prandtl) regimes, differing in the amplitude and random seed of the initial perturbation; 367 GB in all (listed as 342 GB on the dataset card, which counts in units of 2^{30} bytes). Each trajectory has 200 frames, 0.25 simulation time units apart, of the four fields on a 512×128 grid: uniform and periodic in x, and on Chebyshev–Gauss nodes in y, which cluster near the walls to resolve the boundary layers (the files store a uniform placeholder for y; we identified the nodes from the data, Appendix C). We keep The Well's official split, which assigns whole trajectories: 1,400 for training, 175 for validation and 175 for testing (40 / 5 / 5 in each regime), so no frame of a validation or test trajectory is seen in training. The study trains on nested subsets of the training split; this report uses the 10% subset (140 trajectories, four of the 40 in each regime, 29 GB) and evaluates on the full test split. The validation split serves every choice made before testing: the learning-rate search, monitoring during training (64 fixed windows), the frame at which onset is separated from developed convection, and the start frames of the rollouts. The test split is used only for the final evaluation, after these choices were fixed. Fields are normalised with The Well's training statistics.

Task. From four consecutive frames of buoyancy, pressure and both velocity components, a model predicts the next frame; longer forecasts come from feeding its predictions back in (autoregressive rollout). Each trajectory gives 196 one-step windows.

Models. The Well's own implementations, trained from random initialisation, with matched parameter counts (Table 1). All three use GELU activations. For TFNO, released-checkpoint reproduction and training with the current code must be distinguished: the paper-era code skipped TFNO's Fourier layers, an error fixed upstream in November 2025. We scored the released TFNO weights with the forward pass they were trained with, and our own TFNO, trained with the fixed code (The Well at `e63dec8`), uses its Fourier layers (Appendix D).

Method and baselines. The method is a controlled comparison: the three families are trained from scratch with one recipe on the same data and budgets, and compared on accuracy and on cost to train and to use. Two references anchor the comparison:

- *Persistence* predicts that the next frame equals the last input frame. It needs no training and almost no computation, so a model is worth its training cost only where it beats persistence. It is hard to beat where the flow changes slowly, as during onset.
- *The released CNextU-Net* (The Well's, trained on all the training data for 12 hours on an H100) shows what this architecture reaches with ten times the data and a longer, different schedule. It also validates our pipeline (Section 3).

Planned for the final report is a *numerical-solver baseline*. The data were generated with the spectral solver Dedalus [11], and our solver driver is set up as the data's generator was (same equations, boundary conditions and grid; written, not yet run). Restarted from a model's last input frame at the data's own resolution, the solver reproduces the data up to the chaotic growth of tiny differences, so it would show the dataset's own reproducibility limit rather than a fair competitor for surrogates trained for a few hours. The informative comparison is the solver on coarser grids (256×64 and 128×32), which costs less and is less accurate: at what grid and CPU cost does it match each model's accuracy? Comparing against a numerical method at equal accuracy or equal cost follows McGreivy and Hakim [15]. Three cheaper baselines need no GPU training and can be added to the final evaluation: *linear extrapolation* in time $(2\,y_t - y_{t-1})$, which extends persistence by the latest rate of change; *climatology*, the mean training state of the same regime (a vertical profile of each field), the best constant forecast at long lead times; and *dynamic mode decomposition* [16], a linear model fitted by least squares and the classical data-driven baseline in fluid dynamics, which would show how much of the networks' accuracy needs a nonlinear model.

Appendix B: Detailed Training Recipe

Sources of the reference settings: the paper (learning rates, 12-hour H100 budget) and The Well's training configuration at commit `e63dec8`:

- `configs/optimizer/adam.yaml`: AdamW, weight decay 10^{-4} (the paper's text says 0.01; we follow the code);
- `configs/data/rayleigh_benard.yaml`: batch 64;
- `configs/trainer/defaults.yaml`: 200 epochs;
- `configs/lr_scheduler/cosine_with_warmup.yaml` with `train.py`: 5 warmup epochs from 0.1·lr, then cosine to 0.1·lr.

Our software: The Well `e63dec8`, neuraloperator 0.3.0, PyTorch 2.8.0 (CUDA 12.8), Python 3.13; our repository at `f42b2e2` (training) and `4c0d3a8` (AViT resume, evaluation).

	This study	The Well paper / code
Activation	GELU (all three models)	GELU (same model code)
Optimizer	AdamW, $\beta = (0.9, 0.999)$, weight decay 10^{-4}, no clipping	AdamW, weight decay 10^{-4} in the code (0.01 in the text), no clipping
Loss	one-step MSE, normalised units, equal field weights	same
Batch size, precision	64, fp32	64, fp32
Learning rate	10^{-3} for all, from our search over $\{10^{-4}, 5 \cdot 10^{-4}, 10^{-3}\}$	per model from $\{10^{-4}, 5 \cdot 10^{-4}, 10^{-3}, 5 \cdot 10^{-3}, 10^{-2}\}$: $5 \cdot 10^{-4}$ (CNextU-Net), 10^{-4} (TFNO)
Schedule	500 warmup steps, cosine to 0 over the 4-hour budget	5 warmup epochs, cosine to 0.1·lr over 200 epochs, stopped at 12 h
Budget, hardware	1 h and 4 h of active GPU training on one A100	12 h on one H100, best validation epoch
Training data	10% of the trajectories (100% next)	100%

Table 6: Training recipe: ours against the paper's.

Released CNextU-Net's schedule. By our estimate (our A100 throughput, with an H100 at most three times faster), the paper's 12-hour runs completed at most about 25 of their 200 scheduled epochs, so their learning rate was still above 95% of its peak when training stopped, whereas ours decays to zero within the budget.

Validation VRMSE after 20 min	10^{-4}	$5 \cdot 10^{-4}$	10^{-3}
CNextU-Net	0.841	0.469	**0.408**
TFNO	0.603	0.358	**0.302**
AViT	0.646	0.332	**0.257**

Table 7: Learning-rate search.

Training. All models use AdamW, one-step MSE in normalised field values, batch size 64 and fp32, with a peak learning rate of 10^{-3}, 500 warmup steps and cosine decay to zero over four hours of active GPU training on one A100. The one- and four-hour budgets are the study's compute axis. The paper trained for 12 hours on an H100 with its own learning rates and schedule, so ours is a different experiment (Table 6). Checkpoints are saved at one and four hours. The one-hour results are those checkpoints after one hour of the four-hour schedule, with the learning rate still at about 85% of its peak: they show what each model has learned after one hour of training, not the best a schedule ending at one hour could reach. Each configuration is trained with two seeds, which change the initialisation and the order of the training data. All runs share one 10% subset, so the spread between seeds does not include the effect of which four trajectories per regime were drawn, which may be larger. With two seeds, differences within the seed range are not read as rankings.

Learning rate. The paper's rates ($5 \cdot 10^{-4}$ for CNextU-Net, 10^{-4} for TFNO) were chosen for 12-hour runs. With one to four hours a model takes far fewer optimizer steps, and a small learning rate that pays off over a long run may not get far enough in a short one, so we tested whether a higher rate suits our budgets. An equal-budget search (20 minutes of GPU training per run, 10% data, validation only) chose 10^{-3} for all three models (Table 7). At this budget the higher rate was clearly better than the paper's: 0.408 against 0.469 for CNextU-Net at $5 \cdot 10^{-4}$, and 0.302 against 0.603 for TFNO at 10^{-4}. The Well swept $\{10^{-4}, 5 \cdot 10^{-4}, 10^{-3}, 5 \cdot 10^{-3}, 10^{-2}\}$; we tested its lower three rates and omitted the two largest to limit tuning cost. Since 10^{-3} is the largest rate tried, it is the best candidate tested, not an established optimum, and the 20-minute ranking may differ from the ranking after four hours.

Batch size and precision. Batch 64 (The Well's) fits every model in fp32 on an A100 80 GB; CNextU-Net peaks at 64.8 GB of memory reserved by PyTorch (60.9 GB allocated, Table 13 in Appendix H). The current TFNO implementation requires fp32 FFT operations, so all models use fp32 for this comparison. This is an implementation constraint, not a claim that mixed-precision TFNO is impossible. The choice favours TFNO: in the batch-64 profile, bf16 cuts the GPU time of a training step by a factor of 1.7 for CNextU-Net and 2.2 for AViT, and CNextU-Net's memory from 58.7 to 34.5 GB. With mixed precision both would take more optimizer steps in the same budget; AViT would also need faster storage to benefit, since it already waits for data.

Appendix C: Metric Details

VRMSE. For one window (one predicted frame) and one field f, The Well's metric code (its `VRMSE` class at commit `e63dec8`) computes

$$\mathrm{VRMSE}_f = \sqrt{\frac{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_{f,i} - y_{f,i})^2}{s_f^2 + \varepsilon}}, \qquad s_f^2 = \frac{1}{N-1}\sum_{i=1}^{N}(y_{f,i} - \bar{y}_f)^2, \qquad \varepsilon = 10^{-7},$$

over the $N = 512 \times 128$ grid points of the target frame. The four fields (buoyancy b, pressure p, and the velocity components u, w) are averaged with equal weights, and reported values are means over windows with equal weight per window (one-step: all 196 windows of each of the 175 test trajectories; rollouts: per step, over trajectories and start frames). Two variants use the same formula and averaging and differ only in the field values passed in: the *paper protocol* scores normalised fields, $(y_f - \mu_f)/\sigma_f$ with The Well's training statistics ($\sigma_f = 0.249, 0.149, 0.208, 0.180$ for b, p, u, w); our *physical-field VRMSE* scores the fields as stored. How and where the two differ is explained below.

Failure horizon. For a rollout, the first step whose field-mean physical-field VRMSE exceeds a threshold (1, 2 or 5, all reported). A field held constant at the target's spatial mean scores VRMSE ≈ 1, so a forecast above threshold 1 says less about the spatial pattern than that mean alone. Rollouts start at frames 20, 80 and 140 of every test trajectory and run 30 steps. We report the fraction of rollouts that stay below each threshold through step 30 (Table 4) and, for threshold 1, its value at every step (Kaplan–Meier survival; Fig. 6b).

Physical-field VRMSE and the paper protocol. Without ε the two would be identical, since the ratio is invariant to an affine rescaling of each field. With ε, normalising multiplies s_f^2 by $1/\sigma_f^2$ (16–45), so the floor shrinks from 10^{-7} to the equivalent of $\varepsilon\sigma_f^2 \approx 2$–$6 \times 10^{-9}$. During the first $\sim$50 frames the velocity is nearly zero (exactly zero at $t = 0$) and its spatial variance is comparable to 10^{-7}, so the physical-field score is damped there and the normalised one is not. This is the whole difference between the released CNextU-Net's 0.669 (paper protocol) and 0.224 (physical fields): on early windows they score 2.510 and 0.766, while from frame 50 on both give 0.038.

Vertical grid. The files store evenly spaced y coordinates, but the values lie on the 128 Chebyshev–Gauss nodes of the solver's grid. With these nodes the data satisfy the boundary conditions: the Chebyshev interpolant gives $b = 1$ and 0 at the walls to within 6×10^{-10}, the velocity vanishes there to within 9×10^{-8} of its root-mean-square value, and the conductive part of the Nusselt number (the volume-averaged $-\partial b/\partial y$) is 1 to within 3×10^{-9}; evenly spaced and Gauss–Lobatto nodes fail these checks. The spectrum and the Nusselt number use these nodes; the models and VRMSE treat the 128 rows as an image and use no coordinates.

Kinetic-energy spectrum. At each grid row, the two velocity components are Fourier-transformed along the periodic x axis; $E(k_x)$ is the one-sided energy $\frac{1}{2}(|\hat{u}|^2 + |\hat{w}|^2)$ at $k_x = 2\pi n/L_x$ ($L_x = 4$), averaged over the Chebyshev-spaced rows with quadrature weights, so that summing over k_x gives the domain-

mean kinetic energy. Spectra are averaged over the rollouts of each regime (5 trajectories × 3 start frames) at steps 1, 10 and 30, separately for predictions and truth.

Inference time. Measured as a scientist would use a model: one forecast at a time (batch 1), fp32, on one A100 80 GB, with the input already on the GPU. After 10 warm-up predictions we time 50 one-step predictions (reported per step) and one full 30-step autoregressive rollout, with a CUDA synchronisation before each clock stops. At batch 1 the GPU is far from fully used, so this latency includes per-call overhead as well as computation. For many forecasts at once, such as ensembles, we also report the forward-pass time per sample in a batch of 64, measured during profiling (fp32, with gradient tracking as in training, so an upper bound for inference).

Appendix D: TFNO Forward Pass and the Released TFNO

In The Well's code, commit `2d8d7eaf` (2024-06-03, "Added gradient checkpointing") called TFNO's Fourier blocks without keeping their output, so the model computed only its lifting and projection layers; commit `e4301602` (2025-11-17, "actually use the TFNO blocks") restored the assignment (`x =` before the block call) [12]. The paper's TFNO results and the released TFNO checkpoint (uploaded 2025-03-28) date from that period. For the pipeline check we scored the released TFNO weights with the forward pass they were trained with (Fourier blocks skipped): it reproduces the published value, while with the corrected forward pass the same weights score ≈10.5. Our own runs use The Well at commit `e63dec8`, which includes the fix. Upstream issue #49 reports related inconsistencies between released checkpoints and published numbers [14].

Model	Paper-protocol one-step VRMSE (published)	Physical-field one-step all / onset / developed	Rollout from frame 0, steps 6–12 / 13–30
Released TFNO (paper-era forward)	0.663 (0.657)	0.300 / 0.623 / 0.190	142 / 169 (published > 10)
Our TFNO (10% data, 4 h)	0.488	0.209 / 0.502 / 0.109	51 / 77
Persistence	0.202	0.199 / 0.121 / 0.226	0.27 / 0.50

Table 8: The released TFNO, scored by our pipeline on the test split, against our TFNO (seed mean) and persistence.

Why the released TFNO tracks persistence (Fig. 4). With its Fourier blocks skipped, the model is only its lifting and projection layers, which act on each grid point separately: every output pixel is computed from the same pixel's four input frames (and at most its position), with no information from neighbouring pixels. Such a model cannot move structures, such as rising plumes or flow carried by the convection rolls, so what it learns is close to copying each pixel's last value with a small correction from its recent trend: persistence with a little extrapolation. In developed convection it is therefore only slightly better than persistence (physical-field VRMSE 0.190 against 0.226). During onset the fields barely change, so copying is nearly exact, while the model's small per-pixel adjustments are divided by a near-zero spatial variance and become large relative errors. The published all-window value (0.657) hides this, because every model scores poorly on the onset windows. Our TFNO, which uses its Fourier layers, is better than the released TFNO everywhere (developed 0.109 against 0.190; paper protocol 0.488 against 0.663) despite 10% of the data and 4 hours.

Appendix E: Further Pipeline-Check Observations

Three observations shape how we report results:

- **Aggregate relative error is sensitive to the early frames** (Fig. 4). For the first ∼50 frames the fluid is nearly at rest; the evaluated models have large relative velocity errors there, and copying the last frame (persistence) is nearly exact. From frame 50 on, the released CNextU-Net scores 0.038, against an all-window mean of 0.224 in physical fields (0.669 under the paper protocol). Near-zero target variability can amplify relative error, so this does not establish that absolute errors are largest early on. We therefore also report early (*onset*) and later (*developed*) windows separately.
- **Rollout error varies substantially with Rayleigh number.** At Ra = 10^{10} the released CNextU-Net's median over steps 13–30 is 1.7; at 10^6 it is 69.
- **The fixed test evaluation is the clearer reproduction target.** The paper's validation values use 20 random batches per epoch; exact agreement would additionally require the same sampled batches and evaluation state.

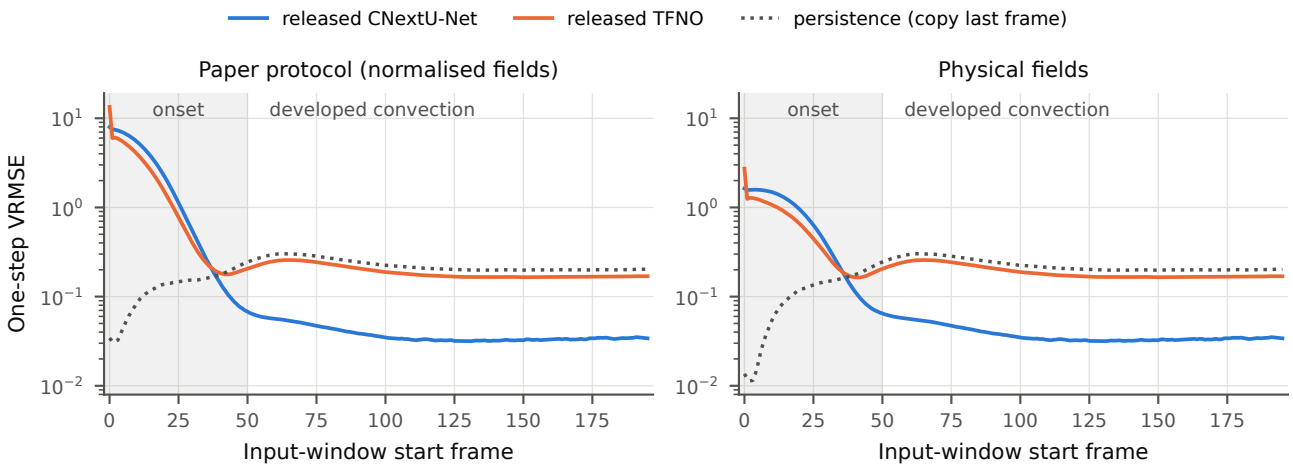


Figure 4: Released checkpoints' one-step VRMSE by window on the test set, mean over the 175 trajectories, under the paper protocol (left) and on physical fields (right). Shaded: windows whose first input frame is below 50, reported as *onset*; the others are *developed*. The cut at frame 50 is operational, chosen where the released checkpoints' error levels off on the validation split; it is not a transition verified for every regime. Persistence copies the last input frame. The released TFNO tracks persistence because of the paper-era bug (Section 3, Appendix D).

Appendix F: Resource Accounting

Resource accounting. *Active GPU training time* is the measured training time counted against the budget (optimizer steps only; the timer is described below). *Allocated GPU time* is the duration for which a job occupies a GPU, including data waits and other overhead. *Service units (SU)* are the cluster's charges on its paid partition. Queue time is reported separately and is not allocated GPU time.

Budget clock. Data are read from the cluster's shared file system, which delivers at most ∼98 training samples per second into one node and stalls for 5–22 minutes at a time, on all nodes at once. TFNO and AViT consume 116 and 330 samples per second of GPU time (Fig. 5), so a wall-clock budget would partly measure our storage. The 1-hour and 4-hour budgets therefore count active GPU training time; waiting for data is logged separately.

Timer. Active training time is measured per optimizer step with a host-side monotonic clock (Python's `perf_counter`), from the moment the data loader returns the batch to just after a CUDA synchronisation (`torch.cuda.synchronize`) following the optimizer update. It includes the host-to-device copy of the batch (pinned memory, non-blocking), setting the learning rate, the forward pass, the loss, the backward

pass, the optimizer step and the host-side launch overhead of these operations. It excludes the wait for the next batch (logged as data-wait time), reading the loss value, logging, validation, checkpoint writes and start-up. After a crash the clock resumes from the last saved checkpoint, so lost steps are not counted. A milestone checkpoint is written after the first step at which the cumulative time reaches 1 h or 4 h (overshoot below one step).

Validation sample. The 64 validation windows are fixed and evenly spaced over the 34,300 one-step windows of the validation split (175 trajectories × 196), taken in the dataset's order (regime, trajectory, time): indices round(linspace(0, 34299, 64)). They cover all 35 regimes (one or two windows each) and 64 different trajectories; 18 of them (28%) are onset windows, close to the 26% of all windows. The learning-rate search used 512 windows chosen the same way (all 35 regimes, all 175 trajectories, 25% onset); the two samples share 8 windows. This monitoring sample does not replace the full test evaluation.

Billing. Paid jobs are billed 32 SU per GPU-hour plus 1 SU per core-hour of allocated time (40 SU per hour for one A100 and 8 cores); the free partition is not billed. Runs that started on the free partition and resumed on the paid one are billed only for the paid part.

Pilot against actual. The pilots predicted 4 h of GPU training to take 6.0 / 8.0 / 9.7 h of wall time (CNextU-Net / TFNO / AViT) at the stall rate observed then; the 10% runs took 5.2–6.0 / 6.9–7.1 / 11.1–12.2 h.

	Validation VRMSE (seed 0 / 1)		Epochs	Allocated time	SU
Model	1 h	4 h	in 4 h	per run	per run
CNextU-Net	1.043 / 0.772	0.167 / 0.178	28	5.2–6.0 h	189–191
TFNO	0.508 / 0.524	0.310 / 0.325	61	6.9–7.1 h	274–286
AViT	0.468 / 0.418	0.161 / 0.167	175	11.1–12.2 h	443–488

Table 9: The 10% runs (140 training trajectories). Validation: 64 fixed windows, physical-field VRMSE. SU covers the paid partition only: CNextU-Net's first attempts ran on the free partition, so its SU corresponds to 4.8 h of its 5.2–6.0 h.

Hours per run	CNextU-Net	TFNO	AViT
10% runs, measured			
Queue wait before training started	0.1–0.9	0.5–0.7	6.7–8.8
GPU training (the budget)	4.0	4.0	4.0
Waiting for data	0.8–1.1	2.5–2.8	6.1–7.3
Start-up, and steps replayed after a preemption or crash[a]	0.13–0.65	0.14–0.15	0.68–0.69
Validation (64 windows every 30 minutes)	0.01–0.02	0.02–0.10	0.18
Checkpoint writes	0.23–0.24	0.08–0.15	0.03–0.13
Allocated time of training	**5.2–6.0**	**6.9–7.1**	**11.1–12.2**
Test evaluation job (1-hour and 4-hour checkpoints)	0.87–0.95	0.57	0.34–0.38
Queue wait before the evaluation started	1.0–14.0	1.9–2.4	3.0–3.3
Training attempts (seed 0 / seed 1)	2 / 2	1 / 1	2 / 2
100% runs, projected			
Waiting for data	1.0–2.0	3.0–4.5	9.5–12.5
Allocated time of training	**5.5–6.5**	**7.5–9**	**14–17**
Test evaluation job	∼0.9	∼0.6	∼0.4

Table 10: Time per run (range over the two seeds). [a]A run that starts once spends about 0.15 h in start-up (environment, data index, model build); the larger values belong to runs with two attempts, which also replayed the steps since their last saved checkpoint. Projection for 100%: the shared file system delivers at most ∼98 samples/s, below TFNO's (116) and AViT's (330) demand, and the full training split (∼290 GB) no longer fits in a node's page cache; start-up, validation, checkpoint writes and evaluation as measured. Queue waits depend on the cluster's load and are not projected.

Component	Allocated GPU-hours
Six 10% runs, completed (24.0 of them active training)	49.8
Learning-rate search, profiling, pipeline check, environment checks, first replication try	18.5
Test evaluation of the six 10% runs and persistence (completed)	4.2
Six 100% runs (projected, 24.0 active; Appendix F)	∼54–65
Test evaluation of the 100% runs	∼4.5
Core study	∼131–142

Table 11: Allocated GPU-hours of the core study: completed parts from Slurm accounting (2026-11-11), the rest projected. No contingency is included; the abstract's ∼100 hours included ∼30, and a similar margin would bring the total to ∼160–170.

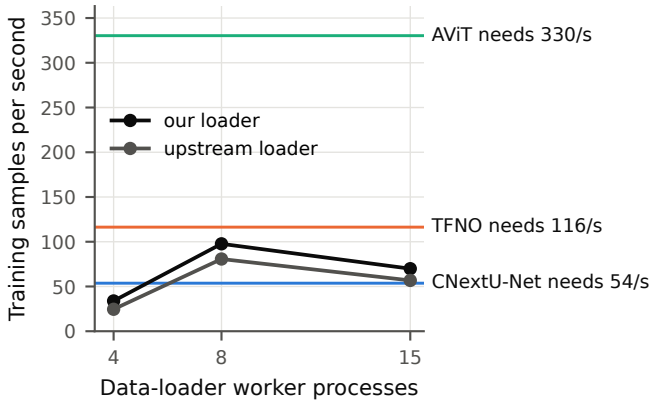


Figure 5: Training samples per second read from the shared file system by one node, against the rate at which each model consumes them on the GPU (horizontal lines).

Appendix G: 10% Results in Detail: Rollouts, Field-Wise Error and Physical Fidelity

The full discussion of the results summarised in Section 5, followed by the rollout error at every step (Fig. 6), the error by field (Table 12) and kinetic-energy spectra (Fig. 7).

Training (Fig. 1). The six runs completed their 4-hour budgets (Table 9). On a fixed validation sample of 64 windows (Appendix F), AViT and CNextU-Net reach similar error after 4 hours; TFNO's loss plateaus after ∼1.5 hours. CNextU-Net's validation error peaks near one hour while its learning rate is still high; this temporal association does not establish the learning rate as the cause.

Test evaluation (all 175 test trajectories, both seeds; Table 3, Fig. 11 in Appendix K). After 4 hours AViT and CNextU-Net have similar one-step error over all windows (0.134 and 0.136) and TFNO a higher one (0.209), but the two phases separate them: in developed convection CNextU-Net is clearly the most accurate (0.050, against 0.075 for AViT and 0.109 for TFNO), consistent with U-Net variants being among the strongest baselines in neural PDE benchmarks [6, 1], while AViT is the most accurate during onset. The across-seed standard deviation of the all-window VRMSE is at most 0.006 (seeds only; the subset is fixed). The released CNextU-Net, trained on all the data for longer, remains better in developed convection (0.038 against 0.050). This gap is not a measure of what ten times more data buys, because the two also differ in hardware, learning rate and schedule (Appendix B). The 100% runs will measure the effect of data with everything else fixed. Persistence is better than every model during onset on the four-field mean, a velocity effect (Table 12), and 2–4.5 times worse in developed convection.

Rollouts (Table 4, Fig. 6). One-step accuracy does not carry over to rollouts in the same order. This is expected for models trained only on one-step targets: in a rollout they receive their own slightly wrong

outputs, inputs they never saw in training [9], and uncontrolled error growth over an autoregressive rollout is a known failure mode [10]. Among our trained models, CNextU-Net is the most stable: after 30 steps from frames 80 and 140 its error (0.53, 0.40) is below persistence's (0.86, 0.54), and 89% of its rollouts stay below VRMSE 1 through step 30. AViT, despite a similar one-step error, degrades fastest (0.80, 0.65; 63%): averaged over the rollouts from frames 80 and 140, its error after 30 steps (0.72) is no better than persistence's (0.70), and from frame 20, during onset, it grows to 5.4. TFNO lies in between. Persistence keeps more rollouts below the thresholds because its error levels off near 0.7, while it is 2–5 times less accurate than every model over the first steps from developed flow (Fig. 6a). Under the paper's frame-0 protocol all three models land in the "> 10" range; CNextU-Net (15 / 30) is below the released CNextU-Net (36 / 35).

With only one hour of training (Tables 3 and 4; the four-hour runs' checkpoints after one hour). The ranking differs from four hours, and no model is best on every count. AViT has the lowest one-step error over all windows (0.39, against 0.47 for TFNO and 0.73 for CNextU-Net) and during onset; in developed convection CNextU-Net (0.081) and AViT (0.085) are close and TFNO trails (0.120). Over rollouts TFNO is the most stable: 60% of its rollouts stay below VRMSE 1 through step 30, against 34% for the other two, and from developed flow it stays more accurate than persistence for the first 21 steps, against 14 for CNextU-Net and 12 for AViT. CNextU-Net, the best model after four hours, is the least reliable after one: its two seeds differ most (all-window VRMSE 0.85 and 0.60), and none of one seed's rollouts stays below VRMSE 1, against 69% for the other. With one hour of training, AViT is the better choice for one-step predictions and TFNO for multi-step forecasts; after 30 steps none of the three is more accurate than persistence.

What the predictions look like (Fig. 2; Ra $= 10^{10}$ in Appendix K). The figure shows buoyancy, while the spectra (Fig. 7) are computed from velocity, so the two are related qualitative evidence rather than measurements of the same field. At Ra $= 10^8$ CNextU-Net follows the true plumes through 30 steps. TFNO blurs them and builds a spurious warm layer along the bottom wall. By step 12 AViT shows rectangular artifacts at the size of its 16×16-pixel patches. At Ra $= 10^{10}$ the true flow is far finer than any of the three reproduces; CNextU-Net keeps the large-scale plume pattern but smooths the small structures.

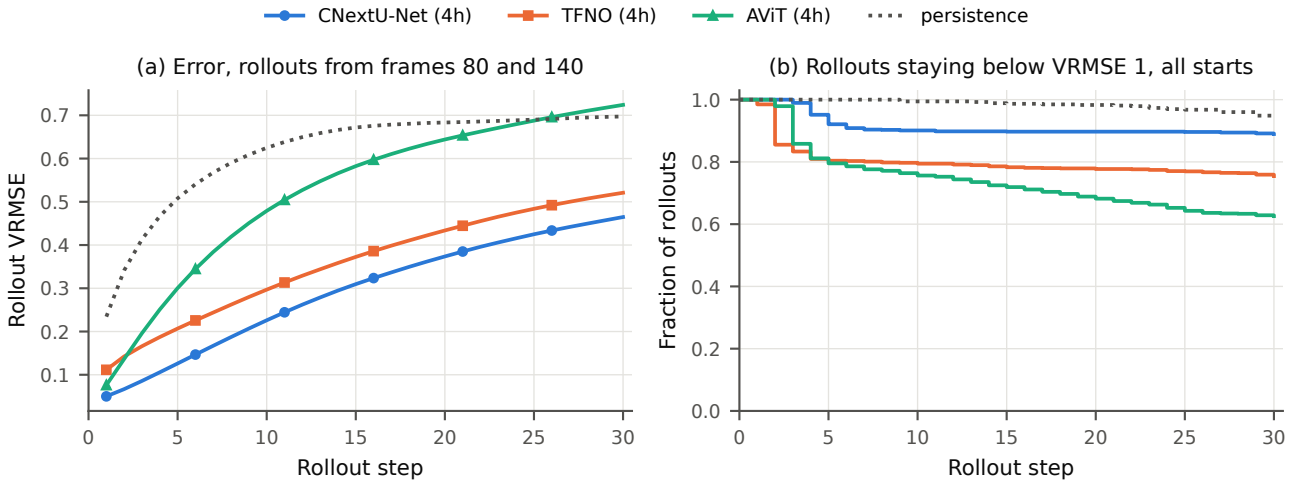


Figure 6: Rollouts of our 10% models at 4 hours (seed mean) and persistence. (a) Physical-field VRMSE at each step, mean over the rollouts from frames 80 and 140 (developed convection). (b) Fraction of all 525 rollouts that have stayed below VRMSE 1 up to each step (Kaplan–Meier survival).

Field-wise error (Table 12). The four-field mean hides where the error lies. In developed convection buoyancy is the hardest field for every model (CNextU-Net: 0.092, against 0.024 for pressure and about 0.04 for each velocity component), consistent with the sharp gradients of its thin plumes and thermal

boundary layers. During onset the picture reverses: buoyancy and pressure errors are small, and the mean comes almost entirely from velocity (CNextU-Net: 0.86 and 0.64), whose near-zero spatial variance magnifies relative error (Section 3). On buoyancy all three models beat persistence even during onset (0.031–0.045 against 0.066), and CNextU-Net and AViT also on pressure, so the models' onset deficit against persistence is a velocity effect. After 30 steps from developed flow, CNextU-Net and TFNO stay below persistence on every field, while AViT is worse than persistence on horizontal velocity (1.06 against 0.93), level on vertical velocity and pressure, and better only on buoyancy. After one hour, CNextU-Net's large all-window error is almost entirely onset velocity (6.8 for u); in developed convection it is best or tied on three of the four fields.

	One step, developed				One step, onset				30 steps from frames 80, 140			
Model	b	p	u	w	b	p	u	w	b	p	u	w
CNextU-Net, 4 h	**0.092**	**0.024**	**0.039**	**0.044**	0.032	0.012	0.863	0.639	**0.384**	**0.293**	**0.598**	**0.584**
TFNO, 4 h	0.170	0.054	0.096	0.116	0.045	0.016	1.164	0.783	0.466	0.307	0.665	0.646
AViT, 4 h	0.119	0.039	0.061	0.080	**0.031**	**0.008**	**0.651**	**0.542**	0.497	0.419	1.059	0.922
CNextU-Net, 1 h	**0.126**	**0.046**	0.079	**0.072**	0.051	0.034	6.829	3.557	1.017	0.729	1.280	1.249
TFNO, 1 h	0.180	0.062	0.108	0.128	0.057	0.029	3.745	2.190	0.925	**0.603**	**0.930**	**0.908**
AViT, 1 h	0.129	**0.046**	**0.074**	0.088	**0.040**	**0.018**	**3.092**	**2.047**	**0.837**	0.652	1.484	1.209
Released CNextU-Net	0.069	0.020	0.030	0.034	0.024	0.011	1.465	1.563	–	–	–	–
Persistence	0.272	0.122	0.245	0.264	0.066	0.014	0.202	0.201	0.531	0.408	0.934	0.916

Table 12: Field-wise test VRMSE (physical fields, mean over the two seeds): buoyancy b, pressure p, horizontal and vertical velocity u, w. Our models: 10% data; the released CNextU-Net: 100%. Bold: best of our models at each budget.

Physical fidelity (4-hour checkpoints; Fig. 7). The kinetic-energy spectrum shows at which scales each model's errors lie. It is a complementary diagnostic, not the paper's binned spectral error, and it describes how energy is distributed over scales, not where structures are: a model can match the spectrum with plumes in the wrong places. The values below are medians over the 35 regimes and both seeds of the predicted-to-true ratio, taken per wavenumber at steps 1, 10 and 30 (the steps evaluated). CNextU-Net stays within 0.80–1.57 below $k_x = 100$ and reaches 190–530 at the highest wavenumbers. TFNO is too smooth: between $k_x \approx 25$ and 100 its ratio falls to 0.33 after one step and 0.09 after 30, and it reaches 9–50 at the highest wavenumbers. AViT stays within 0.97–1.34 below $k_x = 100$ after one step, but after 30 steps it ranges from 0.55 to 21 there and reaches about 5×10^4 at the highest wavenumbers. The TFNO deficit is consistent with its spectral layers keeping only the 16 lowest Fourier modes in each direction ($k_x \lesssim 25$) [5], and AViT's growth with the patch-shaped artifacts in Fig. 2; neither is established as the cause. The excess small-scale energy of CNextU-Net and AViT is typical of surrogates trained with MSE, which weights each scale by its energy and so barely constrains low-energy small scales (the spectral bias of neural networks [7]); such surrogates are known to neglect non-dominant, often high-frequency, components [8]. Persistence repeats its last input field, so its predicted spectrum is fixed within a rollout; changes in its ratio reflect the evolving true spectrum. Ratios at the smallest scales should also be read alongside absolute spectra, because very small true energy can magnify the ratio.

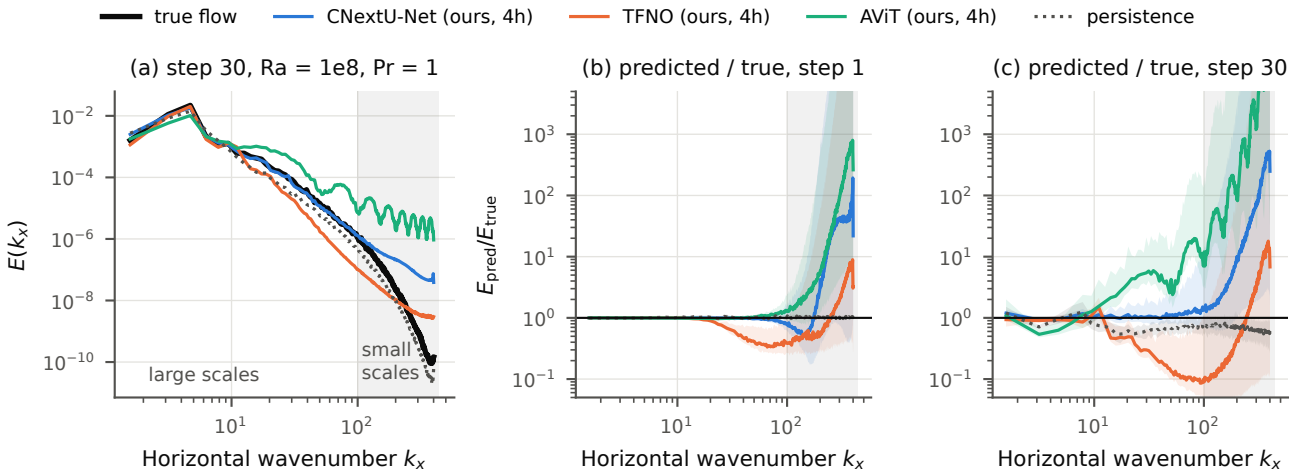


Figure 7: Horizontal kinetic-energy spectra. (a) True and predicted spectrum after 30 rollout steps for one regime ($\mathrm{Ra} = 10^8$, $\mathrm{Pr} = 1$). (b, c) Predicted over true spectrum after 1 and 30 steps, per wavenumber: median over the 35 regimes (band: middle 50%); 1 is exact. Shaded: small scales ($k_x > 100$).

Appendix H: Accuracy Against Cost and Choosing a Model

Accuracy against cost (Fig. 8, Table 13). Because data and compute are the limiting resources, we set each model's accuracy against what it costs to train and to use.

- *Training time* (Fig. 8a). From one to four hours the developed-convection error falls by 38% for CNextU-Net (0.081 to 0.050), 12% for AViT (0.085 to 0.075) and 9% for TFNO (0.120 to 0.109). At one hour CNextU-Net and AViT are close; CNextU-Net pulls ahead only with more training, while TFNO has nearly plateaued.
- *Cost to train* (Fig. 8b). Every run trained for the same 4 hours, but the GPU time it held differed. AViT is the most efficient on the GPU (six times as many optimizer steps as CNextU-Net, a seventh of the memory), but it consumes 330 samples per second, more than our shared storage delivers (∼98; Fig. 5). Each AViT run therefore held a GPU for 11–12 hours, against 6.9–7.1 for TFNO and 5.2–6.0 for CNextU-Net. On faster storage this ranking would change; the demand per GPU-second is the portable number. CNextU-Net's 61 GB of training memory at batch 64 needs an 80 GB GPU or a smaller batch.
- *Cost to use* (Fig. 8c). For a scientist running forecasts, inference is the recurring cost. Every model predicts the next 512×128 frame in 4–8 ms and a 30-step forecast (7.5 time units) in 135–249 ms, using under 0.5 GB of GPU memory. For one forecast at a time TFNO is the fastest; AViT takes 1.2 times and CNextU-Net 1.8 times as long. CNextU-Net's extra 114 ms per forecast buys a 30-step error of 0.47 instead of TFNO's 0.52, while AViT is slower than TFNO and, after 30 steps, no more accurate than persistence. The ranking changes for many forecasts at once: in batches of 64 the forward pass costs 0.89 ms per sample for AViT, 3.15 ms for TFNO and 6.74 ms for CNextU-Net, so for ensembles or many initial conditions AViT is 3.5 times cheaper per forecast than TFNO and 7.6 times cheaper than CNextU-Net. For context, The Well reports that generating the data with the Dedalus solver took about 6,000 to 50,000 s per parameter setting on CPU nodes [13]. That figure covers many trajectories with an unstated parallel layout, so it is not comparable to a per-forecast time; the solver baseline in the plan would measure that comparison directly.

Choosing a model under limited data and compute (Table 14). No family is best on every count. CNextU-Net is the most useful forecaster in developed convection at 10% data and 4 hours, at the price

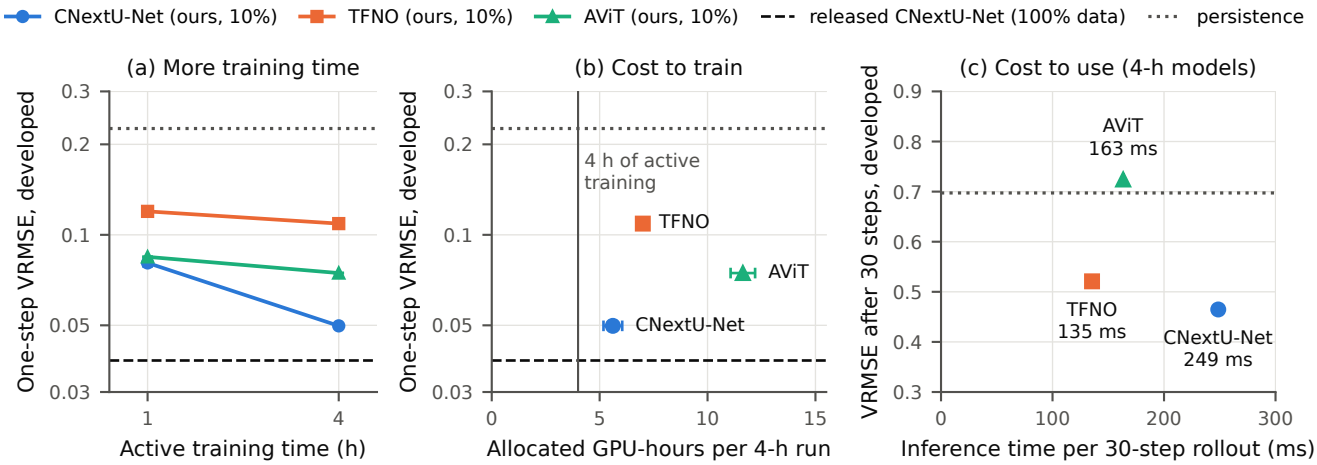


Figure 8: Accuracy against cost, 10% data (seed mean; bars: range over seeds). (a) One-step test VRMSE in developed convection at the 1-hour and 4-hour checkpoints. (b) The same error after 4 hours against the GPU time each run held, including data waits and overhead; vertical line: the 4 hours of active training. (c) Rollout VRMSE after 30 steps from frames 80 and 140 against the time to compute one 30-step forecast (batch 1, A100). Dashed: the released CNextU-Net (100% data); dotted: persistence.

	Training (4-hour budget, batch 64)				Inference			
Model	Optimizer steps	Epochs	Peak memory, alloc. (reserved)	GPU time held per run	per step, batch 1	30 steps, batch 1	per sample, batch 64	Memory, batch 1
CNextU-Net	12,122–12,194	28	60.9 (64.8) GB	5.2–6.0 h	8.3 ms	249 ms	6.74 ms	0.31 GB
TFNO	26,339–26,352	61	48.6 (56.9) GB	6.9–7.1 h	4.4 ms	135 ms	3.15 ms	0.44 GB
AViT	74,978–75,060	175	8.8 (8.9) GB	11.1–12.2 h	5.4 ms	163 ms	0.89 ms	0.23 GB

Table 13: Cost to train and to use, on one A100 80 GB in fp32. Peak training memory: allocated by tensors (reserved by PyTorch's allocator). GPU time held: allocated time of the training job, including data waits, start-up, validation and checkpoints (Appendix F). Per sample, batch 64: forward pass in the training profile (with gradient tracking, an upper bound for inference). Inference memory: peak allocated at batch 1.

of the slowest inference and the most training memory; TFNO is the fastest for single forecasts but too smooth; AViT is the cheapest on the GPU, in training and for batched forecasts, but needs fast storage and does not stay accurate over a rollout.

If the priority is	Best of the three	Trade-off
Accurate forecasts of developed convection that stay stable over 30 steps	CNextU-Net	Slowest inference (249 ms per 30 steps); 61 GB training memory at batch 64; gains most from longer training
Fastest single forecast (batch 1)	TFNO (135 ms per 30 steps)	Too smooth: loses energy between $k_x \approx 25$ and 100; largest one-step error in developed convection; gains least from longer training
Least training memory, most optimizer steps per GPU-hour, cheapest batched forecasts	AViT (8.8 GB; 0.89 ms per sample at batch 64)	Needs storage delivering ∼330 samples/s; rollouts degrade fastest, no better than persistence after 30 steps
Convection onset (first 50 frames)	None: persistence (four-field mean)	Every model is less accurate than copying the last frame on velocity, whose variance is near zero; on buoyancy all three are more accurate
Only one hour of training	AViT (one step); TFNO (rollouts)	AViT: lowest one-step error (0.39); TFNO: 60% of rollouts stay below VRMSE 1, against 34%; CNextU-Net varies most between seeds; none beats persistence after 30 steps

Table 14: Which model to use under which constraint, from the 10% results, after four hours of training unless the row says otherwise. These hold for these configurations and budgets and may change with all the training data.

Appendix I: Additional Analysis

Two analyses go beyond the abstract's metrics: how the error depends on the Rayleigh number, and whether the models reproduce the heat that convection carries across the layer.

Error by Rayleigh number (Fig. 9a, b). Each Rayleigh number groups seven Prandtl numbers (35 test trajectories). The one-step error in developed convection grows four- to tenfold from $\mathrm{Ra} = 10^6$ to 10^{10} (CNextU-Net 0.016 to 0.111, TFNO 0.042 to 0.191, AViT 0.016 to 0.154), as convection becomes more turbulent, with finer plumes and thinner boundary layers; persistence's error grows too (0.098 to 0.351), because the flow changes more between frames. The ranking is the same at every Rayleigh number, except that AViT ties CNextU-Net at 10^6. The released CNextU-Net, trained on all the data, matches ours at 10^6 and 10^7 (0.018 and 0.017, against 0.016 and 0.020) and is better only at higher Rayleigh numbers (0.085 against 0.111 at 10^{10}): the 10% model falls short where the flow is most complex, with the caveat that the two runs also differ in schedule and hardware. After 30 steps from developed flow, CNextU-Net and TFNO stay below persistence at every Rayleigh number, while AViT is worse than persistence up to 10^8 (0.81 against 0.66 at 10^6) and better only at 10^9 and 10^{10}. Over all three start frames, the fraction of CNextU-Net's rollouts that stay below VRMSE 1 instead rises with Rayleigh number (0.67 at 10^6, 0.98 at 10^{10}). This is consistent with convection starting later at low Rayleigh number, so that more rollouts from frame 20 begin in the near-quiescent onset, where relative errors are large; the same effect underlies the released model's Rayleigh-number dependence (Appendix E).

Heat transport (Table 15, Fig. 9c). The Nusselt number is the heat carried across the layer relative to conduction alone. Volume-averaged, and in the units of the data (diffusivity $\kappa = (\mathrm{Ra}\,\mathrm{Pr})^{-1/2}$, unit height and temperature difference), it is $\mathrm{Nu} = 1 + \sqrt{\mathrm{Ra}\,\mathrm{Pr}}\,\langle w\,b\rangle$, computed for every predicted and true frame of each rollout. Two properties shape the comparison. The factor $\sqrt{\mathrm{Ra}\,\mathrm{Pr}}$, 3×10^4 to 3×10^5 at $\mathrm{Ra} = 10^{10}$, amplifies small errors in the correlation between vertical velocity and buoyancy. And the instantaneous value fluctuates strongly: across test trajectories the true Nusselt number one step after frame 80 ranges from −249 to 1,363, so relative errors near zero crossings are unbounded and means are dominated by outliers. We therefore report medians, and the error of the Nusselt number averaged over each 30-step rollout from frames 80 and 140; its sign shows whether a model predicts too much or too little heat transport.

After one step the four-hour CNextU-Net reproduces the instantaneous Nusselt number to within 2%

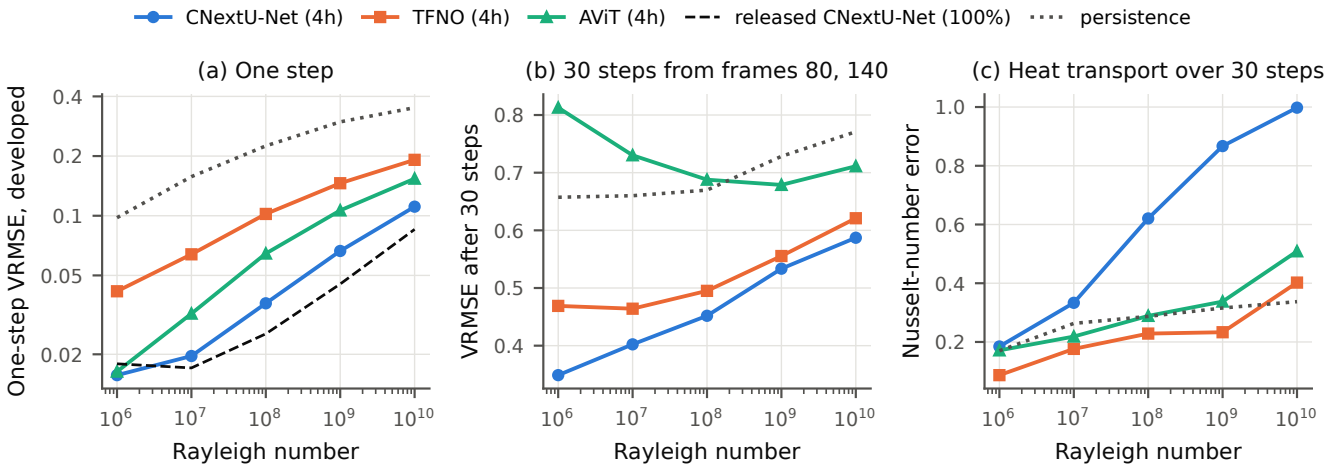


Figure 9: Error by Rayleigh number, four-hour models (seed mean; each point averages seven Prandtl numbers, 35 test trajectories). (a) One-step VRMSE in developed convection. (b) VRMSE after 30 steps from frames 80 and 140. (c) Median relative error of the Nusselt number averaged over those 30-step rollouts. Dashed: released CNextU-Net (100% data); dotted: persistence.

	One step	Averaged over 30 steps	
Model (10% data, checkpoint)	median |error|	median |error|	median signed error
CNextU-Net, 4 h	**0.020**	0.485	−0.015
TFNO, 4 h	0.072	**0.196**	+0.064
AViT, 4 h	0.037	0.275	−0.167
CNextU-Net, 1 h	0.265	5.479	+5.451
TFNO, 1 h	0.285	0.906	+0.205
AViT, 1 h	**0.081**	**0.353**	+0.138
Persistence	0.076	0.267	+0.023

Table 15: Relative error of the volume-averaged Nusselt number on the 350 rollouts from frames 80 and 140 (175 trajectories × 2), median over rollouts, mean over the two seeds. One step: the instantaneous value after the first step. Signed: negative means too little heat transport. Bold: best of our models at each budget.

(median), better than AViT (3.7%), TFNO (7.2%) and persistence (7.6%), and it is the best or tied at every Rayleigh number. Over a 30-step rollout the picture changes. TFNO comes closest to the true heat transport (0.20, against 0.27 for persistence). AViT systematically predicts too little (median −17%, down to −43% at $\mathrm{Ra} = 10^{10}$). CNextU-Net is unbiased on average (−1.5%) but has the largest spread (0.49), which grows with Rayleigh number to 1.0 at 10^{10}. Its excess small-scale energy (Fig. 7) could create spurious correlations between w and b that the factor $\sqrt{\mathrm{Ra}\,\mathrm{Pr}}$ amplifies; this is consistent with the data but not established. After one hour CNextU-Net predicts about six times the true heat transport. Heat transport is therefore largely a negative result: a low one-step VRMSE does not guarantee the right heat flux over a rollout, and the MSE objective does not constrain it.

Appendix J: Issues in Detail

All issues met so far, including those already resolved or handled (Table 16), and where each 10% run's GPU time went (Fig. 10).

Area	Issue	Effect so far	Status
Data on HPC3	The shared file system (`/pub`) delivers at most ∼98 training samples per second into one node and stalls for 5–22 minutes at a time, on all nodes at once	TFNO and AViT need 116 and 330 samples/s; per 4-hour run they waited 2.5–2.8 h and 6.1–7.3 h for data (Fig. 10)	**Open.** Assess node-local staging before the 100% runs, whose 294 GB training split no longer fits in a node's page cache
	File-system errors (`Errno 70`)	Crashed both AViT runs at the same moment on two nodes, and two learning-rate-search runs	**Mitigated.** Runs resume from their last checkpoint
	The upstream loader reopened every file for each sample	The first replication attempt ran at 5.8–47 s per step instead of 0.55–1.42 s	**Resolved.** Our loader keeps files open (same samples bit for bit, 1.2–1.4 times faster) and uses more workers
	The files store a uniform placeholder for the vertical coordinate	Boundary checks, spectra and the Nusselt number would be wrong	**Resolved.** Nodes identified from the data (Appendix C)
Compute on HPC3	The free GPU partition preempts jobs, and its A100s were often held for higher-priority jobs	Three 10% runs were preempted; jobs waited without starting	**Worked around.** Training and evaluation moved to the paid partition (class account)
	Queue waits on the paid partition	Median about 3 h; up to 12 h before training and 15 h before evaluation	**Open.** The 100% runs need about a day of slack
	Allocated time is billed, including data waits	2,160 SU of the shared class allocation so far; the 100% runs need about 2,400 more	**Open.** Falls if staging removes the waits
Training	TFNO's Fourier layers fail in bf16	Every model trains in fp32, which favours TFNO: bf16 would speed up a training step 1.7 times for CNextU-Net and 2.2 times for AViT	**Accepted** for this comparison; stated as a caveat
	The learning rate 10^{-3} is the largest tried, from a 20-minute search	The optimum may be higher; CNextU-Net's validation error peaks near one hour, where its two seeds differ most	**Accepted**; stated as a caveat
Evaluation	VRMSE is ill-conditioned during onset, where the velocity is nearly zero	Persistence beats every model on the four-field mean during onset	**Handled.** Onset and developed convection reported separately, and per field
	Models are trained only on one-step targets	AViT's rollouts are no better than persistence after 30 steps; no model keeps the heat transport right over a rollout	**Open.** Rollout-aware training is outside the study's fixed recipe
	The released TFNO was trained with a bug that skipped its Fourier layers; the paper's validation values cannot be reproduced	Only the released CNextU-Net is a valid reference, compared on the test split	**Documented** (Appendix D)

Table 16: All issues met so far, their effect and their status.

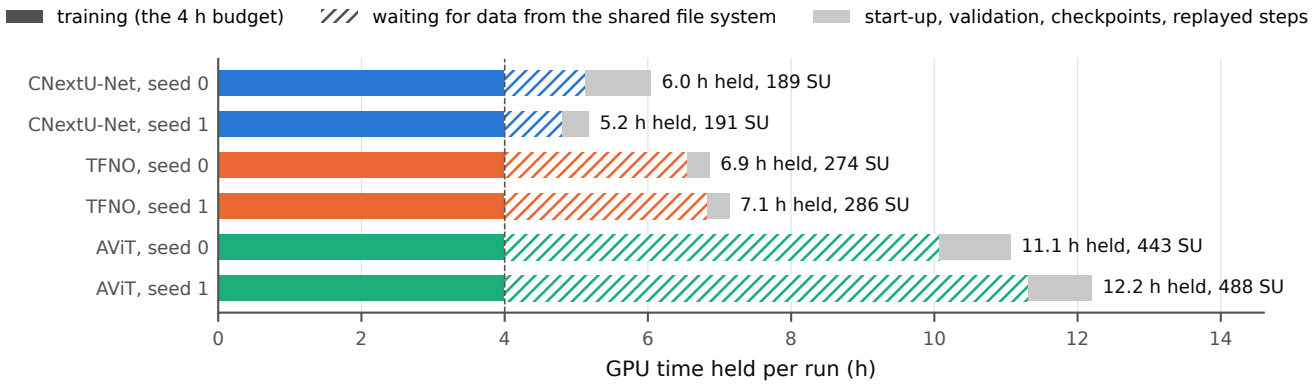


Figure 10: Where each 10% run's GPU time went: active training (the 4-hour budget), waiting for data from the shared file system, and other overhead (start-up, validation, checkpoint writes, and steps replayed after a preemption or crash). SU: billed on the paid partition; the CNextU-Net runs started on the free partition, which is not billed.

Appendix K: Additional Figures

Test error by window (Fig. 11) and flow snapshots at Ra $= 10^{10}$ (Fig. 12).

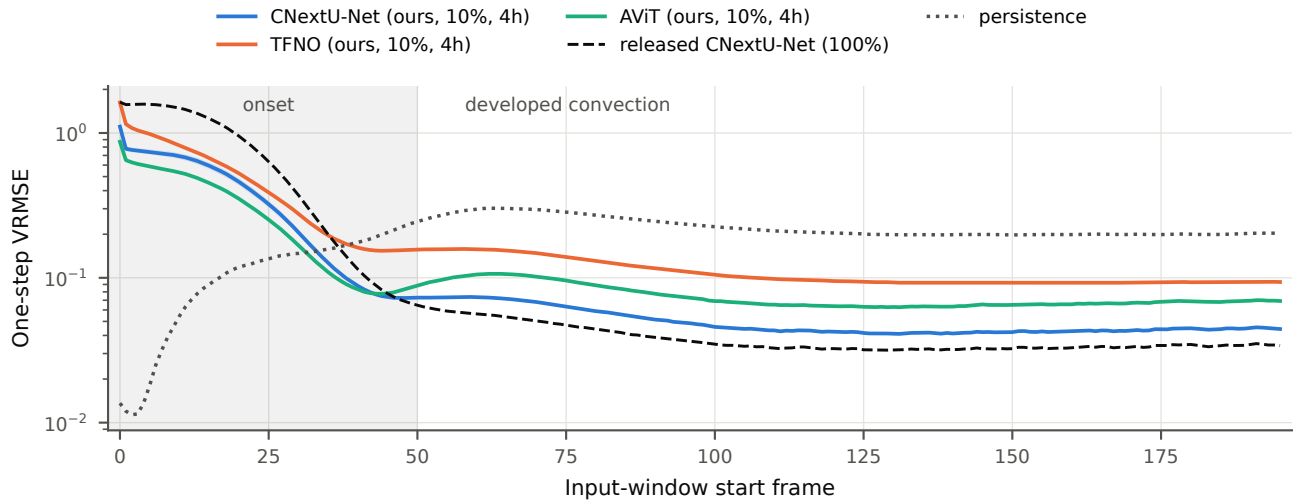


Figure 11: One-step test VRMSE (physical fields) by window, mean over the 175 test trajectories: our 10% models at 4 hours (seed mean; band: range over seeds), the released CNextU-Net trained on all the data, and persistence. Shaded: onset windows (first input frame < 50; see Fig. 4).

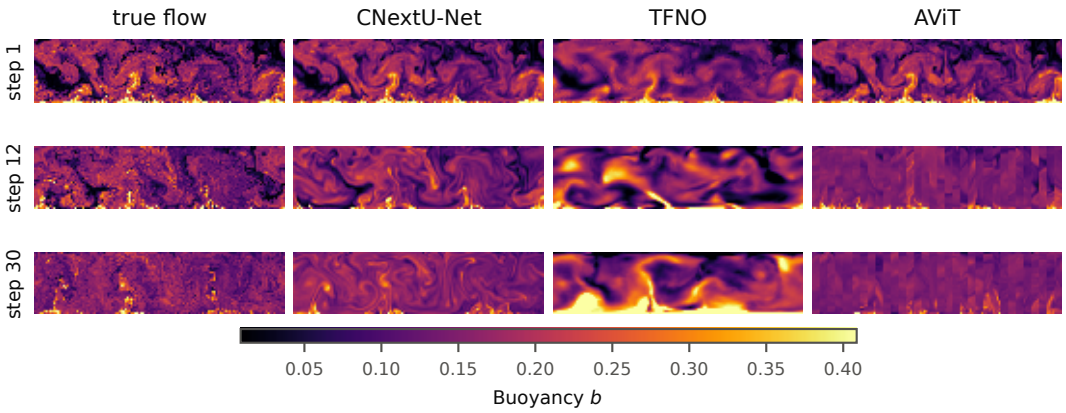


Figure 12: As Fig. 2, at $\mathrm{Ra} = 10^{10}$, $\mathrm{Pr} = 1$.

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https://github.com/PolymathicAI/the_well/commit/2d8d7eaf9eb46b96f215abf39c94b6be4bf9ea5c

https://github.com/PolymathicAI/the_well/commit/e4301602e57f2dddeaa87856dda49bbba47c663b

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https://github.com/PolymathicAI/the_well/blob/e63dec8/datasets/rayleigh_benard/README.md

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